

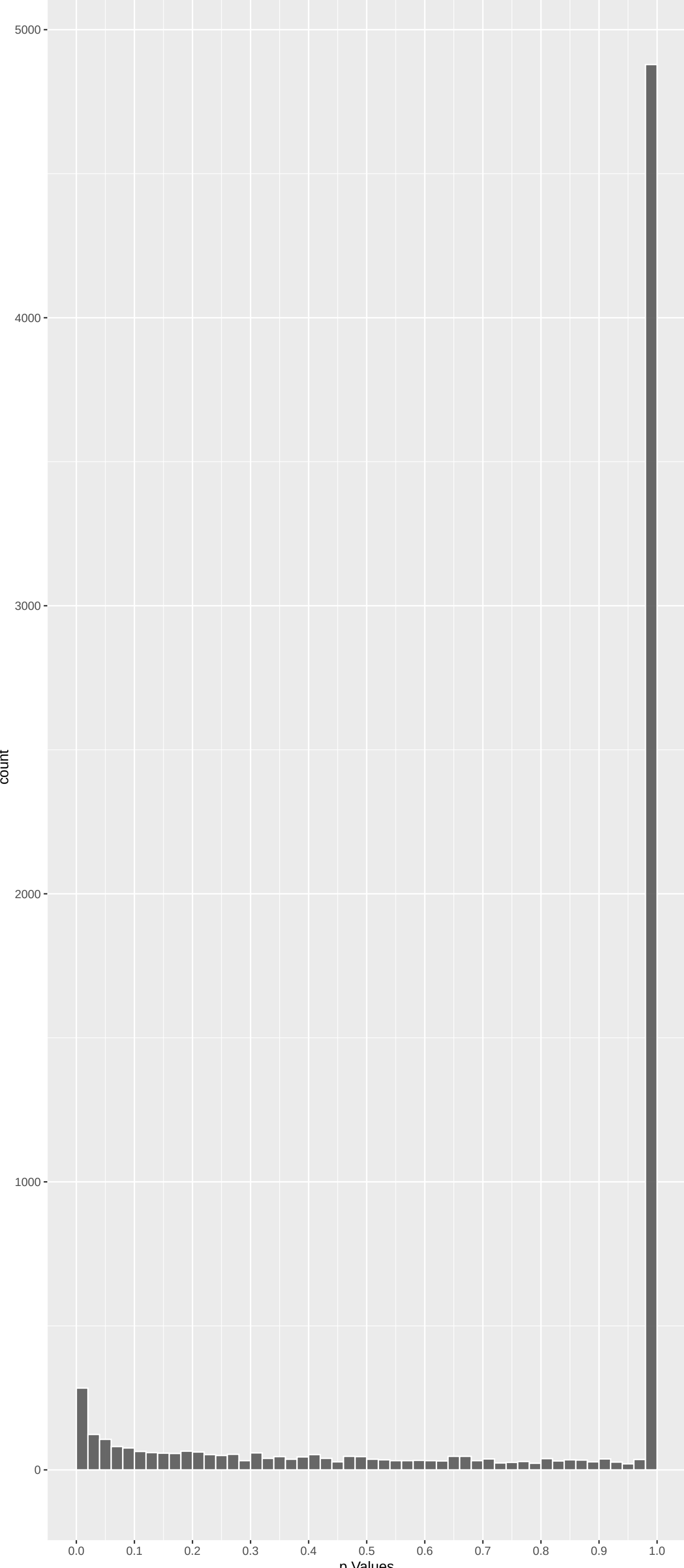
Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
PGBD1	-5.646313	9.835502e-08	1.246e-03	1.000e+00
AFP	5.311204	6.534208e-07	3.749e-03	1.000e+00
MINAR1	-5.255031	8.879988e-07	3.749e-03	1.000e+00
MYH13	-5.148367	1.576580e-06	4.992e-03	1.000e+00
PRSS55	4.942723	4.622340e-06	7.318e-03	1.000e+00
IQGAP3	-4.994909	3.531813e-06	7.318e-03	1.000e+00
CRTC2	-5.024566	3.027427e-06	7.318e-03	1.000e+00
DYNC2H1	-4.955558	4.327379e-06	7.318e-03	1.000e+00
SLC22A16	4.918452	5.233887e-06	7.365e-03	1.000e+00
BPIFB1	4.821028	8.569238e-06	1.085e-02	1.000e+00
RIOX1	4.781892	1.041916e-05	1.200e-02	1.000e+00
MBD1	4.753307	1.200698e-05	1.267e-02	1.000e+00
NOL9	4.674188	1.770711e-05	1.602e-02	1.000e+00
ATP8B2	-4.687261	1.661314e-05	1.602e-02	1.000e+00
IL1RL2	4.620665	2.295073e-05	1.938e-02	1.000e+00
TENM3	-4.597072	2.570820e-05	2.035e-02	1.000e+00
CEMP	-4.550538	3.210534e-05	2.079e-02	1.000e+00
CAPN8	-4.545043	3.295448e-05	2.079e-02	1.000e+00
PIGM	-4.577147	2.828160e-05	2.079e-02	1.000e+00
C1orf141	4.535511	3.447856e-05	2.079e-02	1.000e+00
IQANK1	-4.546661	3.270226e-05	2.079e-02	1.000e+00
HIRA	4.519646	3.716590e-05	2.140e-02	1.000e+00
TACR2	-4.501459	4.049314e-05	2.230e-02	1.000e+00
BNC1	-4.479300	4.493286e-05	2.371e-02	1.000e+00
SPINT4	4.470091	4.691185e-05	2.377e-02	1.000e+00

Top genes by Q-Value non-permuted

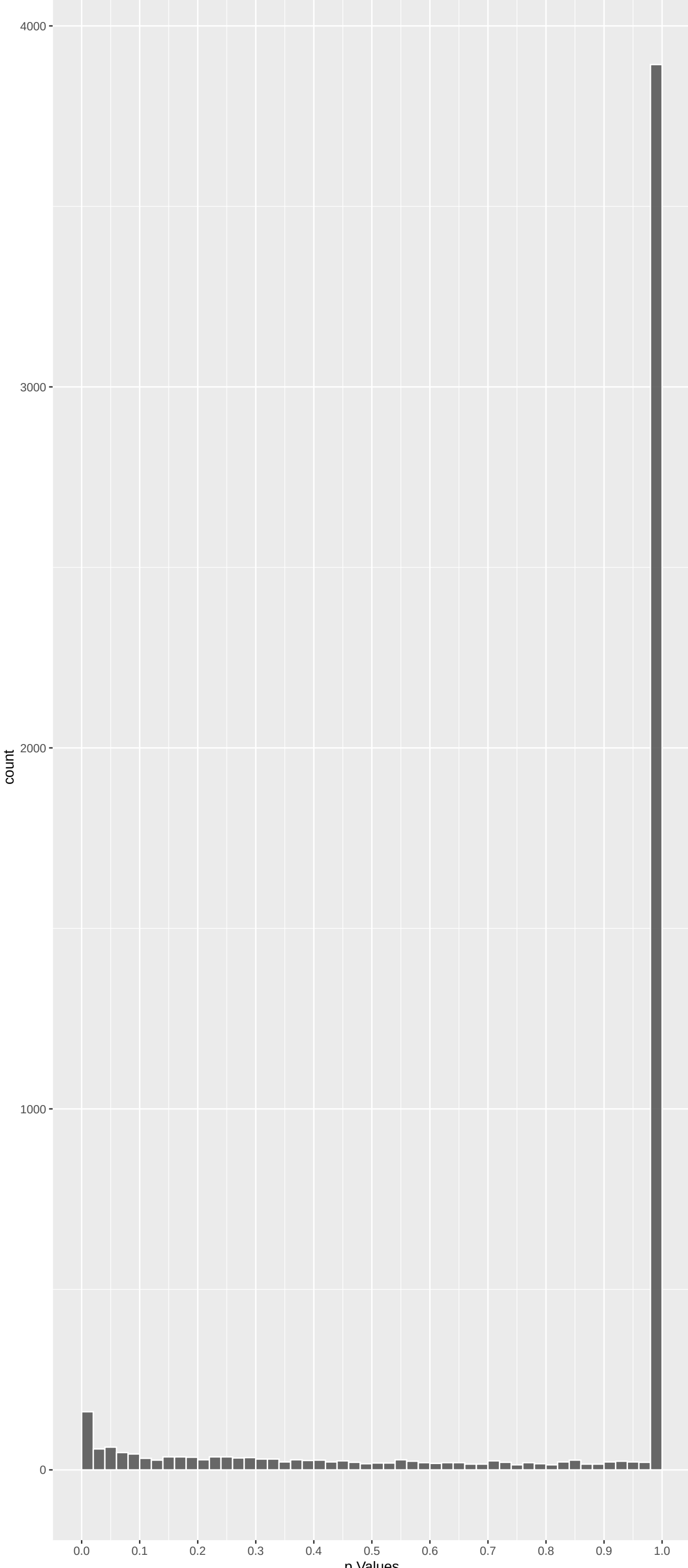
Gene	Rho	P	p.adj	qValueNoperm
M6PR	-0.2190166	1.00000000	1.000e+00	1.000e+00
FKBP4	1.7679711	0.46239443	1.000e+00	1.000e+00
NDUFAF7	-0.3873102	1.00000000	1.000e+00	1.000e+00
FUCA2	-0.5027470	1.00000000	1.000e+00	1.000e+00
DBNDD1	-1.0081884	1.00000000	1.000e+00	1.000e+00
HS3ST1	1.3366056	1.00000000	1.000e+00	1.000e+00
SEMA3F	1.9980208	0.27428643	1.000e+00	1.000e+00
CFTR	-0.5787058	1.00000000	1.000e+00	1.000e+00
CYP51A1	2.7598339	0.03469845	7.551e-01	1.000e+00
USP28	-1.2274013	1.00000000	1.000e+00	1.000e+00
SLC7A2	0.8873768	1.00000000	1.000e+00	1.000e+00
PK4	0.6355754	1.00000000	1.000e+00	1.000e+00
USH1C	-0.2343433	1.00000000	1.000e+00	1.000e+00
RALA	0.0000000	1.00000000	1.000e+00	1.000e+00
BAIAP2L1	1.0532552	1.00000000	1.000e+00	1.000e+00
CACNG3	-1.1020803	1.00000000	1.000e+00	1.000e+00
TMEM132A	-0.7757284	1.00000000	1.000e+00	1.000e+00
DVL2	-1.0945541	1.00000000	1.000e+00	1.000e+00
RPAP3	-0.3867167	1.00000000	1.000e+00	1.000e+00
TRAPP6A	0.5024468	1.00000000	1.000e+00	1.000e+00
SPATA20	-0.1808349	1.00000000	1.000e+00	1.000e+00
CD79B	1.2443971	1.00000000	1.000e+00	1.000e+00
RHBD2	1.4728844	0.84469322	1.000e+00	1.000e+00
RPUSD1	0.5693757	1.00000000	1.000e+00	1.000e+00
TSR3	1.0705882	1.00000000	1.000e+00	1.000e+00

Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
AFP	5.311204	6.534208e-07	3.749e-03	1.000e+00
PRSS55	4.942723	4.622340e-06	7.318e-03	1.000e+00
SLC22A16	4.918452	5.233887e-06	7.365e-03	1.000e+00
BPIFB1	4.821028	8.569238e-06	1.085e-02	1.000e+00
RIOX1	4.781892	1.041916e-05	1.200e-02	1.000e+00
MBD1	4.753307	1.200698e-05	1.267e-02	1.000e+00
NOL9	4.674188	1.770711e-05	1.602e-02	1.000e+00
IL1RL2	4.620665	2.295073e-05	1.938e-02	1.000e+00
C1orf141	4.535511	3.447856e-05	2.079e-02	1.000e+00
HIRA	4.519646	3.716590e-05	2.140e-02	1.000e+00

Gene	Rho	P	p.adj	qValueNoperm
PGBD1	-5.646313	9.835502e-08	1.246e-03	1.000e+00
MINAR1	-5.255031	8.879988e-07	3.749e-03	1.000e+00
MYH13	-5.148367	1.576580e-06	4.992e-03	1.000e+00
IQGAP3	-4.994909	3.531813e-06	7.318e-03	1.000e+00
CRTC2	-5.024566	3.027427e-06	7.318e-03	1.000e+00
DYNC2H1	-4.955558	4.327379e-06	7.318e-03	1.000e+00
ATP8B2	-4.687261	1.661314e-05	1.602e-02	1.000e+00
TENM3	-4.597072	2.570820e-05	2.035e-02	1.000e+00
CEMP	-4.550538	3.210534e-05	2.079e-02	1.000e+00
CAPN8	-4.545043	3.295448e-05	2.079e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Steroid Metabolic Process (GO:0008202)	0.15944633	56	3.767e-05	2.022e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 CYP51A1:375 HSD17B3:409 CYP21A2:412
'De Novo' AMP Biosynthetic Process (GO:0	-0.15293445	4	2.895e-01	1.580e-01	PFAS:128 PAICS:5819 ATIC:5819 ADSL1:5819 NA NA
'De Novo' Post-Translational Protein Fold	0.09781599	15	1.898e-01	1.580e-01	SDFZ1:1887 DNAJB13:1097 HSPA14:1742 GAK:1896 ERO1A:2443 ENTDP5:6847
2-Oxoglutarate Metabolic Process (GO:000	0.15259031	13	5.685e-02	1.580e-01	GOT2:166 TAT:169 OGDHL:185 GOT1:484 L2HGDH:1540 IDH1:6847
3'-UTR-mediated mRNA Destabilization (GO	-0.02404704	11	7.825e-01	1.580e-01	DND1:1427 TRIM71:5819 RC3H1:5819 ZFP361:5819 ZFP361:25819 MOV10:5819
3'-UTR-mediated mRNA Stabilization (GO:0	0.07242310	11	4.057e-01	1.580e-01	MAPK14:1303 HNRNPC:1409 HNRNP0:1754 DAZL:2202 ZFP36:6847 TARDBP:6847
3'-Phosphoadenosine 5'-Phosphosulfate Me	0.24757905	8	1.532e-02	1.580e-01	SULT1A2:97 SULT1B1:964 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 PAPSS2:6847
5S Class rRNA Transcription By RNA Polym	-0.04055920	5	7.535e-01	1.580e-01	GTF3C2:5819 GTF3C3:5819 GTF3C4:5819 GTF3C5:5819 GTF3C6:5819 NA
7-Methylguanosine RNA Capping (GO:000945	0.20867202	2	3.067e-01	1.580e-01	RNMT:569 CMTR2:6847 NA NA NA NA
7-Methylguanosine mRNA Capping (GO:00063	0.20867202	2	3.067e-01	1.580e-01	RNMT:569 CMTR2:6847 NA NA NA NA
ADP Transport (GO:0015866)	-0.02679134	5	8.356e-01	1.580e-01	SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 SLC25A41:12017 NA
AMP Biosynthetic Process (GO:0006167)	-0.18068332	6	1.250e-01	1.580e-01	PFAS:128 APT8:855 PAICS:5819 ATIC:5819 ADSL1:5819 NUDT2:5819
ATF6-mediated Unfolded Protein Response	0.05763797	5	6.554e-01	1.580e-01	CREBZF:656 MBTPS1:6847 XBP1:6847 ATF6B:6847 DDIT3:6847 NA
ATP Biosynthetic Process (GO:0006754)	0.02728418	13	7.335e-01	1.580e-01	TGFB1:780 ATP5F1C:1803 NUOT2:6847 VPSD91:5819 FOLH1:5819 ATP5PF:6847 ATP5PD:6847
ATP Metabolic Process (GO:0046034)	0.02569101	25	6.212e-01	1.580e-01	TGFB1:780 ENPP3:1053 BAD:1081 MYH6:1785 ATP5F1C:1803 PARG:2089
ATP Synthesis Coupled Electron Transport	0.11562172	3	4.879e-01	1.580e-01	NDUF86:949 NDUF5F:6847 NDUFV3:6847 NA NA NA
ATP Transport (GO:0015867)	-0.08235327	9	9.323e-01	1.580e-01	CALHM1:191 SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 CALHM3:5819
AV Node Cell To Bundle Of His Cell Commu	0.04574441	4	7.514e-01	1.580e-01	GJA5:2469 SCN5A:6847 CXADR:6847 GJC3:6847 NA NA
Arp2/3 Complex-Mediated Actin Nucleation	-0.09664430	10	2.900e-01	1.580e-01	WHAMM:727 ARPC5L:796 WASHC1:1166 ARPC1A:1319 ARPC5:5819 ARPC3:5819
B Cell Activation (GO:0042113)	0.06407234	61	8.399e-02	1.580e-01	CD86:34 HDAC9:214 IFNK:316 IL4:679 CASP8:1036 CR2:1071
B Cell Chemotaxis (GO:0035754)	0.05649964	5	6.617e-01	1.580e-01	CYP7B1:734 PIK3CD:6847 HSD3B7:6847 CXCL13:6847 GAS6:6847 NA
B Cell Differentiation (GO:0030183)	0.05405464	44	2.153e-01	1.580e-01	HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAM1:1120 PTPN2:1201
B Cell Homeostasis (GO:0001782)	0.03584262	7	7.426e-01	1.580e-01	TNFRSF13B:400 MEFC2:1121 GAPT:6847 DOCK10:6847 TNFAIP3:6847 TNFAIP3:6847
B Cell Proliferation (GO:0042100)	0.07943161	15	2.870e-01	1.580e-01	IFNK:316 CR2:1071 MEFC2:1121 PTPRC:2183 IL10:6847 CD70:6847
B Cell Receptor Signaling Pathway (GO:00	0.07446488	26	1.890e-01	1.580e-01	SH2B2:544 CTLA4:865 CD38:1042 MEFC2:1121 RFTN1:2025 PTPRC:2183
BMP Signaling Pathway (GO:0030509)	-0.04139601	42	3.542e-01	1.580e-01	LEFTY2:341 LEFTY1:489 GDF10:1012 FAM83G:1081 INHBE:1119 SLC39A5:1205
C-terminal Protein Amino Acid Modificati	-0.08813263	8	3.880e-01	1.580e-01	AGBL4:781 GPLD1:967 AGBL5:5819 AGTPBP1:5819 FOLH1:5819 ATG16L1:5819
C-terminal Protein Deglutamylation (GO:0	-0.13997713	4	3.323e-01	1.580e-01	AGBL4:781 AGBL5:5819 AGTPBP1:5819 FOLH1:5819 NA NA
C-terminal Protein Lipidation (GO:0006050	-0.03478644	3	8.347e-01	1.580e-01	GPLD1:967 ATG16L1:5819 ATG7:10875 NA NA NA
CD4-dicarboxylate Transport (GO:0015740)	0.11620399	10	2.033e-01	1.580e-01	SLC1A1:216 SLC25A12:618 SLC25A18:5819 SLC2A2:1451 SLC25A13:6847 SLC13A5:6847
CD4-positive, Alpha-Beta T Cell Activatio	-0.13020632	5	1.033e-01	1.580e-01	NKG7:147 STOML2:5819 TNFSF4:5819 RSAD2:5819 HMGCB1:5819 NA
CD40 Signaling Pathway (GO:0023035)	0.03632786	7	7.393e-01	1.580e-01	CD86:34 CD40LG:6847 ITGB1:6847 PHB2:6847 TNIP2:6847 ITGAS:6847
CENP-A Containing Chromatin Assembly (GO	0.04334790	5	7.371e-01	1.580e-01	NASP:1553 MIS18A:6847 CENPF:6847 HUJRP:6847 OIP5:6847 NA
COP1 Coating Of Golgi Vesicle (GO:004820	-0.05684820	2	7.807e-01	1.580e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COP1-coated Vesicle Budding (GO:0035964)	-0.05684820	2	7.807e-01	1.580e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COP1 Vesicle Coating (GO:0048208)	-0.03438454	14	6.561e-01	1.580e-01	KLHL12:475 TRAPPCC12:1193 TRAPP3:5819 TRAPPCC2L:5819 TRAPPCC1:5819 TRAPPCC2:5819
COP1-coated Vesicle Budding (GO:0090114	-0.01150596	29	8.303e-01	1.580e-01	TBC1D20:311 KLHL12:475 SH202A:491 TRAPPCC12:1193 TRAPP3:5819 TRAPPCC2L:5819
CRD-mediated mRNA Stabilization (GO:0070	-0.19756075	5	1.261e-01	1.580e-01	IGF2BP1:496 YBX1:1195 DHX9:5819 PABP3:5819 HNRNP0:5819 NA
Cdc42 Protein Signal Transduction (GO:00	-0.04055702	6	7.308e-01	1.580e-01	SHTN1:5819 WAS:5819 RHOV:5819 NTN1:5819 RHOBTB1:5819 RHOBTB2:5819
DNA Alkylation (GO:0006305)	0.09262387	15	2.144e-01	1.580e-01	DNMT3B:31 GATAD2A:146 BAZ2A:343 HLLS:1649 KMT2E:6847 MTRR:6847

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	-0.29109711	27	1.661e-07	1.075e-03	ADAM15:45 PBXIP1:80 MTX1:85 RUSC1:89 RORC:177 FDP5:188
NIKOLSKY_BREAST_CANCER_2Q012_Q13_AMPLICON	-0.14502570	100	5.637e-07	1.825e-03	COL20A1:30 HELZ2:88 MTG2:97 CDH26:104 FAM217B:130 LAMA5:149
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.13697150	92	5.790e-06	1.250e-02	SULT1A2:97 AKR1D1:117 MGST1:161 CYP2F1:176 COMT:179 CYP11A1:201
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.23916671	26	2.444e-05	3.957e-02	HSD17B2:14 HSD11B1:107 AKR1D1:117 COMT:179 CYP11A1:201 HSD17B3:409
REACTOME_BIOLOGICAL_OXIDATIONS	0.09916207	123	1.503e-04	1.947e-01	ALDH1A1:73 SULT1A2:97 HPGDS:136 MGST1:161 CYP2F1:176 COMT:179
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.19383617	28	3.868e-04	2.784e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 SERPINA6:585
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.10671382	95	3.319e-04	2.784e-01	SLC36A1:71 SLC36A2:135 SLC12A3:193 SLC1A1:216 SLC6A6:251 SLC7A6:272
REACTOME_METABOLISM_OF_LIPIDS	0.04463701	576	2.908e-04	2.784e-01	HSD17B2:14 SLC01B1:82 PTGR1:96 HSD11B1:107 AKR1D1:117 HPGDS:136
CAIRO_HEPATOBlastOMA_DN	0.07440518	196	3.441e-04	2.784e-01	TMEM131L:35 AKR1D1:117 SLC46A3:125 CA2:152 SLC27A5:177 OGDHL:185
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.07628784	180	4.324e-04	2.800e-01	IFNARI:58 1LIR2:66 TNF:87 TNFSF10:103 TNFSF15:112 IL1RAP:304
NIKOLSKY_BREAST_CANCER_8Q23_Q24_AMPLICON	-0.09228699	111	7.998e-04	3.176e-01	PLEC:24 FAM83H:202 GPR2:254 TRAPPCC1:286 OPLAH:293 SCRI8:348
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.43146604	5	8.338e-04	3.176e-01	CA2:152 CA4:253 AQP1:762 SLC4A1:1060 RHAG:2146 NA
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.06778672	219	5.733e-04	3.176e-01	SLC22A16:3 SLC36A1:71 SLC01B1:82 SLC10A6:83 SLC36A2:135 SLC12A3:193
WP_2Q12_COPY_NUMBER_VARIATION_SYNDROME	0.17941272	30	6.742e-04	3.176e-01	CNNM4:163 ARID5A:424 CNNM3:508 KANSL3:580 COL2A1:605 DUSP2:677
WP_STEROID_HORMONE_PRECURSOR_BIOSYNTHESI	0.36775507	7	7.532e-04	3.176e-01	AKR1D1:117 CYP11A1:201 CYP21A2:412 CYP17A1:883 HSD3B2:1180 SRD5A2:2125
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20027634	24	6.854e-04	3.176e-01	GOT2:166 AGXT:186 CAD:245 IL41:373 GLS2:427 GOT1:484
KEGG_CYSTINE_AND_METHIONINE_METABOLISM	0.21228425	21	7.602e-04	3.176e-01	DNMT3B:31 DNMT1:113 GOT2:166 TAT:169 IL41:373 LDHC:408
NIKOLSKY_BREAST_CANCER_17Q21_Q25_AMPLICON	-0.06055546	248	1.075e-03	3.783e-01	FTS3J:58 HOXB13:67 LRRC59:71 ITGBA:100 ENGASE:119 EVPL:121
WP_CLASSICAL_PATHWAY_OF_STEROIDOGENESIS	0.27189643	12	1.110e-03	3.783e-01	HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 CYP17A1:883 HSD3B2:1180
BENPORATH_ES_WITH_H3K27ME3	-0.03573796	730	1.177e-03	3.810e-01	CEMP10 BNC1:14 PLEC:24 SUSP4:34 ADAM15:45 CAMK2B:46
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	-0.14706730	40	1.297e-03	3.998e-01	PIEZO1:56 CPNE7:138 ZC3H18:408 ANKRD11:431 SPATA2L:545 MVD:604
BENPORATH_SUZ12_TARGETS	-0.03630485	681	1.422e-03	4.187e-01	CEMP10 BNC1:14 PLEC:24 SUSP4:34 ADAMTSL3:50 HOXB13:67
REACTOME_GLUCCORTICOID_BIOSYNTHESIS	0.36870547	6	1.762e-03	4.226e-01	HSD11B1:107 CYP21A2:412 SERPINA6:585 CYP17A1:883 HSD3B2:1180 HSD11B2:6847
WP_ALLOGRAFT_REJECTION	0.11271893	65	1.691e-03	4.226e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXP3:324 ABCB1:397 IL1A:608
WP_PROXIMAL_TUBULE_TRANSPORT	0.13659419	44	1.732e-03	4.226e-01	ATP6VOA4:43 SLC36A2:135 CA2:152 SLC1A1:216 CA4:253 ABCB1:397
KEGG_SULFUR_METABOLISM	0.30192728	9	1.710e-03	4.226e-01	SULT1A2:97 SULT1A4:370 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 CHST13:2118
KEGG_TYPE_1_DIABETES_MELLITUS	0.18085786	25	1.753e-03	4.226e-01	CD86:34 TNF:87 HLA-DQB1:137 IL1A:608 CD80:787 IFNG:1016
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.04988831	333	1.866e-03	4.316e-01	CHRN82:26 SHE:40 CAMK2B:46 HTR1D:55 KCHN2:81 YBX2:87
REACTOME_ARACHIDONIC_ACID_METABOLISM	0.14099083	40	2.044e-03	4.564e-01	PTGR1:96 HPGDS:136 GGT1:470 GPX4:500 CYP1A1:528 GGT5:642
REACTOME_CYTOKINE_BIND_ARRANGED_BY_SUB	0.14731339	35	2.573e-03	5.375e-01	CYP2F1:176 CYP11A1:201 CYP51A1:375 CYP21A2:412 CYP27A1:499 CYP1A1:528
KEGG_GRAFT_VERSUS_HOST_DISEASE	0.21133524	17	2.559e-03	5.375e-01	CD86:34 TNF:87 HLA-DQB1:137 IL1A:608 CD80:787 IFNG:1016
PID_NFAT_TFPATHWAY	0.14661319	35	2.697e-03	5.458e-01	TNF:87 FOXP3:324 DGKA:353 EGR4:646 IL4:679 CTLA4:865
REACTOME_MEIOTIC_RECOMBINATION	-0.15454536	21	2.912e-03	5.714e-01	MRE11:406 ATM:434 H2AZ2:626 MLH1:818 H3-4:831 H2BC9:1058
REACTOME_TNFS_BIND_THEIR_PHYSIOLOGICAL_R	0.17270343	23	3.271e-03	6.186e-01	TNFSF15:112 TNFRSF13B:400 TNFSF14:613 TNFSF8:708 TNFRSF9:836 FASLG:1216
PYEON_HP_V_POSITIVE_TUMORS_UP	0.0915087	73	3.436e-03	6.186e-01	TASOR:41 TAF7L:154 UHRF1:155 TRIM56:411 DTL:423 GLS2:427
KEGG_ALLOGRAFT_REJECTION	0.18445481	21	3.436e-03	6.186e-01	CD86:34 TNF:87 HLA-DQB1:137 ITGB1:6847 TNF:87 IFNG:1016
LASTOWSKA_NEUROBLASTOMA_COPY_NUMBER_UP	-0.07130891	138	3.908e-03	6.214e-01	MINK1:51 FTS3J:58 LRRC59:71 NEURL4:200 CHRN81:265 P4HB:270
BIOCARTA_HBX_PATHWAY	-0.36844491	5	4.327e-03	6.214e-01	SHC1:353 CREB1:565 LAMTOR5:675 SOS1:955 PTK2B:5819 NA
IVANOVA_HEMATOPOIESIS_MATURE_CELL	0.05934998	203	3.673e-03	6.214e-01	ST3GAL5:84 IL33:127 GCNT1:129 TSPAN8:167 TM4SF4:195 COL1A1:209
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	-0.16105691	26	4.487e-03	6.214e-01	TP53BP2:38 ATM:434 CASP2:465 AIFM2:528 TP73:610 PERP:828

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Dermatitis	0.10566782	138	1.950e-05	6.349e-02	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Leprosy	0.13214844	88	1.908e-05	6.349e-02	TNF:87 TNFSF15:112 HLA-DQB1:137 FOXP3:324 IL18RAP:416 CD163:524
Malignant neoplasm of colon stage IV	0.23651636	29	1.347e-05	6.349e-02	PROM1:22 TNFSF10:103 TPX2:286 SMPD1:606 FBLN1:633 VEGFA:756
Hepatitis C	0.05525428	498	3.112e-05	7.599e-02	AFP:1 CLU:17 GGTLC3:29 STAT2:38 IFNARI:58 IL1R2:66
Severe Dengue	0.19085259	38	4.749e-05	9.275e-02	RFXO1:36 TNF:87 TLR7:238 VEGFA:756 CTLA4:865 CXCL8:1051
Cystic Fibrosis	0.06058923	373	6.930e-05	9.668e-02	AFP:1 BP1FB1:4 DNMT3B:31 CELA3B:42 TRPV5:55 TNF:87
Systemic Scleroderma	0.06286578	347	6.705e-05	9.668e-02	PRSS55:2 IFNARI:58 ENG:64 TNF:87 TNFSF10:103 DNMT1:113
Psoriasis	0.05100856	522	8.434e-05	1.030e-01	IL1RL2:8 STAT2:38 TNF:87 TNFSF10:103 ITGAX:104 TNFSF15:112
Balkan Nephropathy	0.35334846	10	1.094e-04	1.188e-01	TNF:87 GSTK1:235 ABCB1:397 ALB:530 SLC06A1:599 IL1A:608
Goldenhar Syndrome	-0.36617357	9	1.427e-04	1.188e-01	TSC1:59 FRAS1:279 TBC1D20:311 FGF8:316 GRIP1:538 CHD7:768
Hepatitis	0.07592066	201	2.231e-04	1.676e-01	AFP:1 PROM1:22 SLC01B1:82 TNF:87 TNFSF10:103 AKR1D1:117
IGA Glomerulonephritis	0.08276435	169	2.192e-04	1.676e-01	CD86:34 TNF:87 ITGAX:104 MAN2B1:106 HORMAD2:118 FCGR2:149
Inflammatory dermatosis	0.10099228	115	1.913e-04	1.676e-01	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Renal salt wasting	0.28288815	14	2.484e-04	1.733e-01	CYP11A1:201 CYP21A2:412 SCNN1A:465 CLCNKA:664 SCNN1G:1157 HSD3B2:1180
Kidney Failure, Chronic	0.05924546	310	3.741e-04	1.857e-01	TRPV1:13 CLU:17 CHDH:77 TNF:87 HSD11B1:107 MSR1:115
Acquired aplastic anemia	0.17905557	34	3.056e-04	1.857e-01	TNF:87 HPGDS:136 GSTK1:235 FOXP3:324 HNF1A:559 SLC06A1:599
Anemia	0.05613830	344	3.926e-04	1.857e-01	DNMT3B:31 ATP2B4:33 IFNARI:58 ENG:64 TNF:87 TNFSF10:103
Helminthiasis	0.23432224	20	2.875e-04	1.857e-01	PRSS55:2 ITGAX:104 IL33:127 AGXT:186 FOXO3:324 IL4:679
Nonalcoholic Steatohepatitis	0.08576194	146	3.657e-04	1.857e-01	DNMT3B:31 TNF:87 TNFSF10:103 HSD11B1:107 DNMT1:113 IL33:127
Polymorphous light eruption	0.39168099	7	3.327e-04	1.857e-01	TNF:87 HPGDS:136 GSTK1:235 SLC06A1:599 IL1A:608 ICAM1:1079
Rare Diseases	0.26410762	15	3.992e-04	1.857e-01	DNMT3B:31 AGXT:186 FGFRI:480 TTC7A:837 CPS1:962 FGF23:1181
Carcinoma, Transitional Cell	0.06323810	260	4.892e-04	1.978e-01	TNF:87 TNFSF10:103 ENG:64 TNF:87 TNFSF10:103 DNMT1:113 HPGDS:136
Brucellosis	0.19170889	28	4.497e-04	1.978e-01	RLSD1:51 TNF:87 PDE4:130 MEVFP:362 IL1A:608 IL4:679
Chronic Kidney Diseases	0.06288187	263	4.897e-04	1.978e-01	CHDH:77 TNF:87 HPGDS:136 AGXT:186 GSTK1:235 NPHS1:242
Dengue Fever	0.10010563	101	5.253e-04	1.978e-01	IFNARI:58 TNF:87 TNFSF10:103 ITGAX:104 NCL2:210 DAXX:315
Hypocalcemia	0.37841226	7	5.266e-04	1.978e-01	TRPV5:55 SLC12A3:193 HNF1A:559 CASRN:720 PTH:899 CLCNKB:1477
Encephalomyelitis	0.06318748	254	5.723e-04	2.054e-01	TRPV1:13 CLU:17 CD86:34 TNF:87 TNFSF10:103 HPGDS:136
Septicemia	0.05618789	322	5.889e-04	2.054e-01	CD86:34 TRPV5:55 RELB:62 TNF:87 IL33:127 PLAUR:164
Alcoholic Liver Diseases	0.13097573	57	6.363e-04	2.094e-01	TRPV1:13 NCAN:15 ALDH1A1:73 TNF:87 HPGDS:136 GSTK1:235
Hepatocarcinogenesis	0.05546298	326	6.433e-04	2.094e-01	AFP:1 CLU:17 PROM1:22 DNMT3B:31 TNF:87 PTHR1:96
Congenital dyserythropoietic anemia	0.28201144	12	7.197e-04	2.117e-01	MAN2A1:108 SEC23B:225 LPIN2:442 GATA1:482 AQP1:762 KLF1:990
Hyponatremia	0.17336229	32	6.947e-04	2.117e-01	COMT:177 CYP11A1:201 ABCB1:397 SCNN1A:465 CLCNKA:664 SLC26A3:665
Sepsis	0.05209496	368	6.779e-04	2.117e-01	CD86:34 ATP6V0A4:43 TRPV5:55 RELB:62 TNF:87 IL33:127
Steroid-sensitive nephrotic syndrome	0.32496165	9	7.370e-04	2.117e-01	TNF:87 PLA2G7:654 VEGFA:756 AEC3:981 CXCL8:1051 IL12B:1171
Dermatitis, Atopic	0.05535255	316	7.939e-04	2.215e-01	RELB:62 ALDH1A1:73 TNF:87 DNMT1:113 IL33:127 HPGDS:136
Immune thrombocytopenic purpura	0.04866606	130	8.925e-04	2.421e-01	DNMT3B:31 CD86:34 TNF:87 TNFSF10:103 IL33:127 FCGR2:149
Acanthocytosis	0.29915179	10	1.056e-03	2.423e-01	KXMS4 VPS31A:206 CAD:245 GATA1:482 KLC:709 PANK2:969
Bacterial Infections	0.07199530	178	9.740e-04	2.423e-01	TNF:87 SLC27A5:177 MPEGL1:188 CPD:223 NF2L2:305 GDE1:256
Diarrhea	0.06205069	237	1.075e-03	2.423e-01	DNMT3B:31 SCNN1A:37 TNF:87 AKR1D1:117 SLC12A3:193 OPT1A:393
Female hypogonadism syndrome	-0.20257727	22	1.010e-03	2.423e-01	FGF8:316 ANOS1:321 ATM:434 PROKR2:436 WDR11:615 GNRHR:620